Active sensing explains cortical variability and improves visual information

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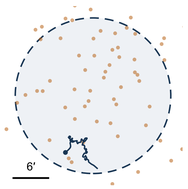
Fixational eye movements: noise or signal?

Even during fixation the eyes drift and make microsaccades, so each repeat of an image lands on a different part of the foveal receptive fields.

Noise view
A repeated stimulus is a repeated input, so trial-to-trial variability is noise to be averaged away.

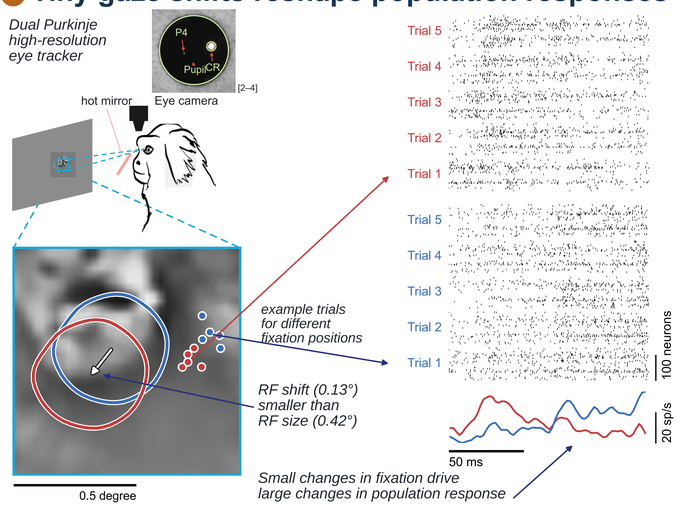
Active-sensing view
Eye motion turns space into time and is part of how the fovea encodes fine detail[5,6].

Q1 How much V1 "noise" do fixational eye movements drive? → 1–5
Q2 Does that activity carry visual information? → 6–8



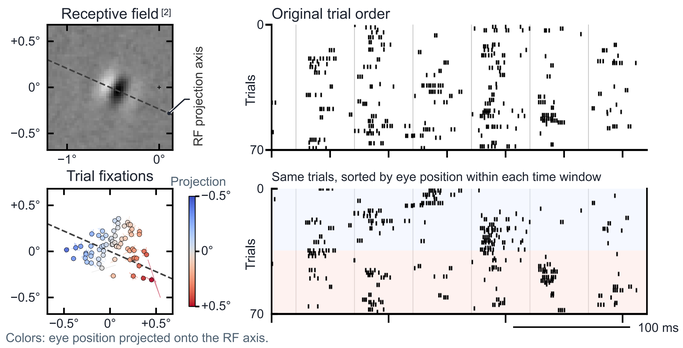
Median gaze on 88 repeats (navy: one path) vs a typical foveal V1 RF (0.42°)

1 Tiny gaze shifts reshape population responses



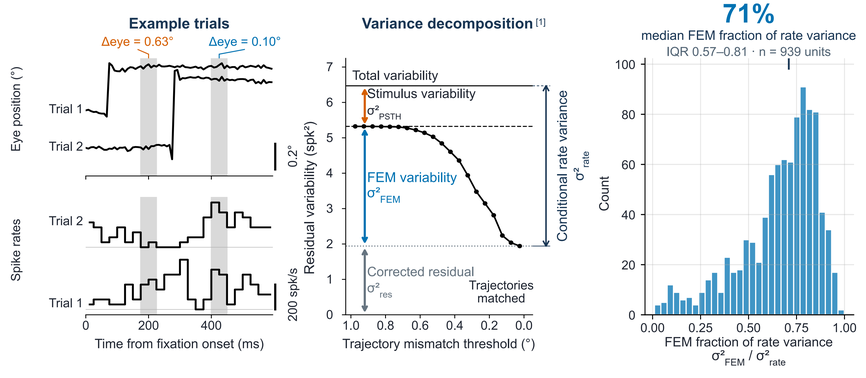
One session, 132 neurons, 5 trials per gaze group, 258–392 ms after fixation onset. Dots: median gaze. RF: population footprint (noise mapping), located from these trials. Dataset: 2 marmosets · 1,022 foveal V1 units · 19 sessions · 20 Hz natural images.

2 One neuron's "noise" tracks the eye

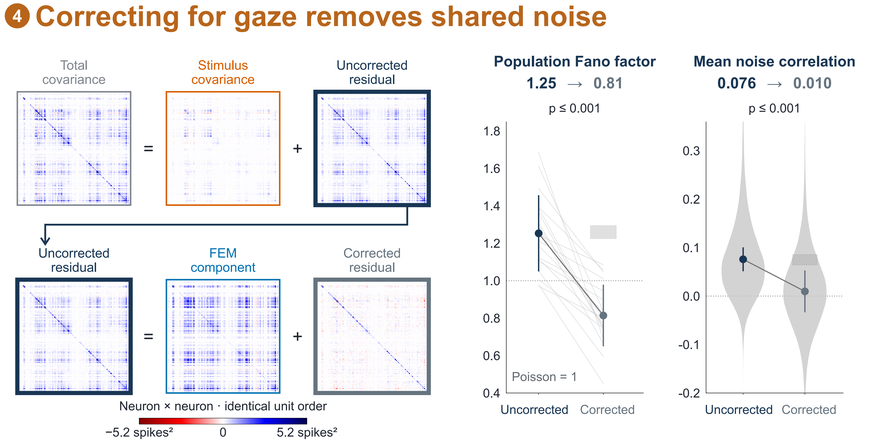


Colors: eye position projected onto the RF axis.

3 Eye movements drive most rate variance (71%)

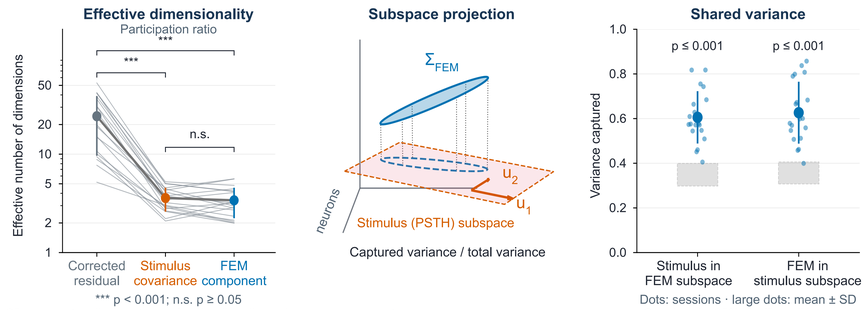


4 Correcting for gaze removes shared noise



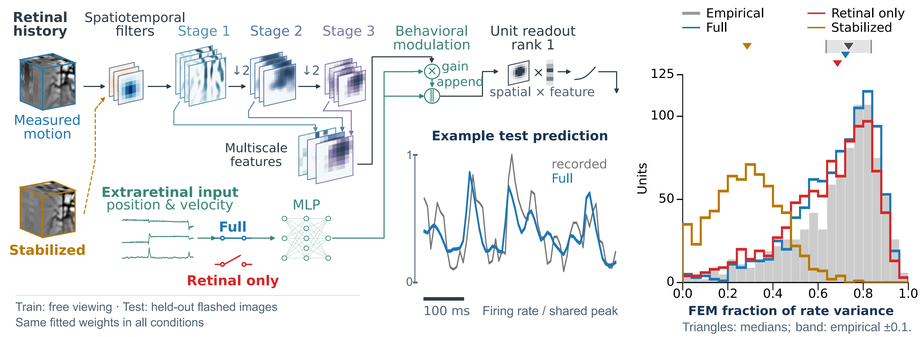
19 sessions; 25-ms bins. Points: mean ± SD. Gray bands: 95% shuffle null; violins: pairwise correlations.
Corrected mean noise correlation: 95% CI −0.010 to 0.029. Both tests compare correction with shuffle.

5 Eye-driven variability follows stimulus dimensions



*** p < 0.001; n.s. p ≥ 0.05
25-ms counts; alignment uses four leading dimensions per component.
Dots: sessions · large dots: mean ± SD
Gray bands: 95% shuffle-null intervals

6 The retinal image, not an eye signal, predicts single trials



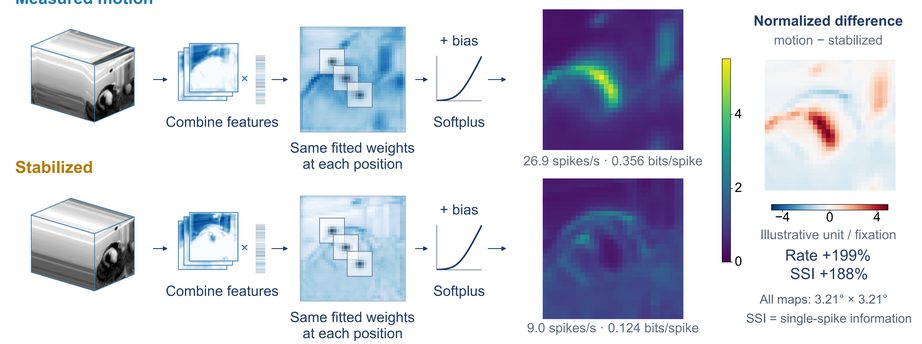
Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions

Triangles: medians; band: empirical ±0.1.

Median fraction of conditional rate variance explained
962 units · 19 sessions

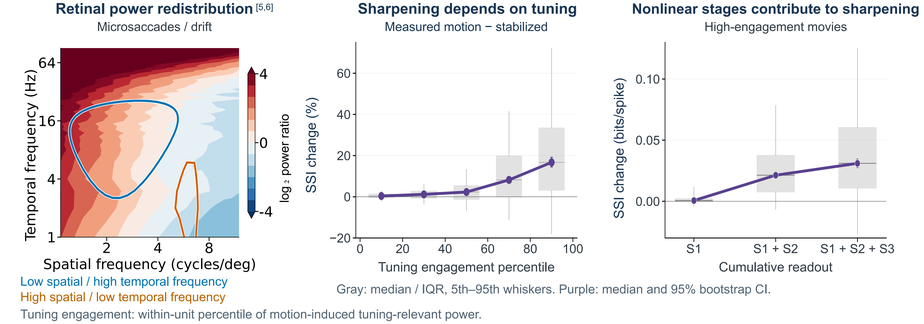
Trial average	Full	Retinal only	Stabilized
0.139	0.235	0.226	0.042

7 In the model, eye motion sharpens spatial coding



8 Sharpening grows with tuning and model depth

725 units



Tuning engagement: within-unit percentile of motion-induced tuning-relevant power.
All quintiles/stages > 0 (Holm-adjusted p < 0.05).
Gray: median / IQR, 5th–95th whiskers. Purple: median and 95% bootstrap CI.

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012

Funding
NEI K99-EY037839-01 (D.P.R.); R01-EY030998 (J.F.M.); R01-EY018363 (M.R.).
Fuster Endowment (A.C.H.); NEI K99/R00-EY032179 (J.L.Y.); NIH/NEI T32-EY007043 (T.K.).

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